

Panasa Ramanna Peddireddi

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PROFILE SUMMARY

As an AWS Cloud Intern transitioning to full-stack development, I design and build scalable backend architectures and distributed systems using Python, Django, and React. I recently shipped a multi-tenant SaaS analytics platform featuring complete PostgreSQL data isolation and engineered a high-performance URL shortener using Redis Streams for asynchronous processing. My hands-on experience with cloud deployment and containerization allows me to bridge the gap between rapid prototyping and robust, production-ready systems.

EDUCATION

Bachelor's Degree - Computer Engineering

Aug, 2023 to Present

Aditya Engineering College

Diploma - Computer Engineering

Dec, 2020 to Jun, 2023

Andhra Polytechnic College

SKILLS & TOOLS

Api Development | Concurrency | Performance Optimization | State Management | System Design | Backend Development | Frontend Development | Communication | Database Management | Docker Compose Configuration | Exponential Backoff Implementation | Hmac Signature Generation | Sdk Development | Solution Architecture | System Reliability | Webhook Implementation | Distributed Systems | Error Handling | Idempotency | Integration Testing | Unit Testing | Automated Testing | Observability | Devops | Software Engineering | Seo | Container Orchestration | Data Modeling | Component Based Architecture | Animation | Responsive Design | Uiux Design | Source Control | Cloud Computing | Queues | Serverless | Infrastructure As Code | Caching Strategies | Cloud Deployment | Data Structures And Algorithms | Progamming | Scalability | System Architecture | Troubleshooting | Data Streaming | Api Integration | Javascript Programming | Prompt Engineering | Eda | Content Creation | Data Management | User Acceptance Testing | Programming | Access Control Patterns | Gas Optimization | Invariant Design | Blockchain | Benchmarking | Database Design | Authentication And Authorization | Full Stack Development | 3d Secure Authentication | Multi Tenancy Architecture | Docker | Javascript | Nodejs | Expressjs | Bull | Postgresql | React | Redis | Vite | Apache Kafka | Express Js | Mysql | Node Js | Zookeeper | Jest | Axios | Nextjs | Tailwind Css | Prismjs | Framer | Git | Localstack | Serverless Framework | Amazon Api Gateway | Aws Lambda | Supertest | Npm | Github | Jaeger | Opentelemetry | Python | Fastapi | Openai | Amazon Sqs | S3 | Rabbitmq | Llama | Ollama | Pg Cursor | Python Csv | Ethersjs | Hardhat | Solidity | K6 | Ajv | Django | Nginx | Express | Go | Rust | Amazon S3 | Java

EXPERIENCE

Technical Hub

May, 2025 to Jun, 2025

AWS Cloud Intern, Internship

[On Site](#)

State Board of Technical Education and Training

Jan, 2023 to Jun, 2023

Frontend Development Intern, Internship

[On Site](#)

PROJECTS

Build a Multi-Tenant Analytics Dashboard with OAuth, Django, React, Redis, and PostgreSQL

[GitHub](#) | [Video](#)

2026-05

- Engineered a **multi-tenant architecture** using **Django** and **PostgreSQL** to guarantee strict data isolation and secure workspace switching.
- Developed a responsive analytics dashboard using **React** and **Tailwind CSS**, integrating secure **OAuth** authentication for multi-tenant workspace management.
- Optimized API query response times using **Redis** caching for KPI metrics and containerized the system with **Docker Compose** and **Nginx**.

Stack: Api Development, Authentication And Authorization, Caching Strategies, Database Design, Full Stack Development, Programming, State Management, System Design, 3d Secure Authentication, Multi Tenancy Architecture, Django, Docker, Postgresql, Python, React, Redis

Build a Distributed URL Shortener with Collision-Resistant ID Generation and Analytics

[GitHub](#) | [Video](#)

- Engineered a scalable distributed URL shortener using **Node.js** and **Express**, featuring a collision-resistant ID generation system backed by **PostgreSQL**.
- Designed a low-latency redirect system utilizing **Redis** for efficient **caching strategies** to minimize database load and optimize application performance.
- Developed an event-driven analytics pipeline using **Redis Streams** and **Docker Compose** to process user redirect data asynchronously and reliably.

Stack: Backend Development, System Design, Api Development, Caching Strategies, Database Management, Benchmarking, Nodejs, Postgresql, Redis, Docker, K6, Express

Build a Real-Time SSE Sports Score Feed Multiplexer

[GitHub](#) | [Video](#) 2026-05

- Engineered a real-time sports score aggregator using **Node.js** and **Server-Sent Events** to stream live updates to concurrent clients.
- Developed reliable event replay and subscription features using **Node.js** and **Express.js** to manage real-time client **state synchronization**.
- Architected fully containerized application environments using **Docker** and **Docker Compose** to simplify local **deployment** and system configuration.

Stack: Api Development, Concurrency, Performance Optimization, State Management, System Design, Backend Development, Frontend Development, Docker, Go, Javascript, Nodejs, Python, Redis, Rust, Expressjs

Build an Event-Driven Asynchronous Image Processing Service with AWS SQS

[GitHub](#) | [Video](#) 2026-05

- Engineered a scalable backend service using **Node.js** and **Express** to ingest and process high-volume user image uploads through **Asynchronous APIs**.
- Architected a decoupled processing pipeline utilizing **AWS SQS** and **AWS S3** to manage message queuing and object storage for **Distributed Systems**.
- Deployed a **Containerized Worker** service with **Docker** and **LocalStack** to locally simulate cloud infrastructure and execute heavy image-processing tasks.

Stack: Backend Development, Distributed Systems, Api Development, Cloud Computing, Automated Testing, Nodejs, Expressjs, Docker, Amazon Sqs, Amazon S3, Localstack, Python, Java

Build a Virtual Network Simulator for Packet Tracing

[GitHub](#) | [Video](#) 2025-12

- Engineered a stateless **REST API** using **Node.js** and **Express.js** to simulate complex network routing and longest prefix matching.
- Programmed custom **DNS resolution** and ordered **firewall rules** directly in **Node.js** without using external networking dependencies.
- Containerized the stateless **backend application** using **Docker** to run and test virtual network scenarios defined in **JSON** configurations.

Stack: Backend Development, Devops, Nodejs, Expressjs, Docker